

Brian Wonjun Lee

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EDUCATION

University of Pennsylvania

MSE in Computer Graphics & Game Technology (CGGT), Submatriculation

BA in Computer Science & Design (Double Major), Benjamin Franklin Scholars

Relevant Coursework: Interactive Computer Graphics, Computer Animation, 3-D Computer Modeling, Introduction to Algorithms, Introduction to Computer Systems, Programming Languages & Techniques

Philadelphia, PA

Expected May 2028

Expected May 2027

EXPERIENCE

Software Engineer Intern, Aleph Lab (Y Combinator)

June 2026 – Aug 2026 · San Francisco (Remote)

- **Stopped a release that would have shipped to zero users** — the planned over-the-air update pointed at a runtime a dependency upgrade had invalidated: it would have deployed, reported success and reached nobody. Caught it against the live binaries' fingerprints and rerouted to a store build
- **Unblocked a population that could never have been tested** — the rollout flag was scoped by email, which cannot resolve before sign-in, so signed-out and first-time users could never receive the new experience, including the app's own login and first-class screens
- **Gave the company a deploy-free way to undo its entire visual layer** — reverting a UI regression used to mean a store release and days of review; it is now one switch at any percentage, over a tested byte-identical off-path
- **Stopped the app telling parents their child had done nothing when the network failed** — two dashboard screens drew a failed request identically to an empty one, so a parent's read on their own kid came from a dropped request
- **Turned a channel that went dark whenever nobody approved a send into one that runs itself** — proposing, clearing its own guardrails and reaching **91% of targeted devices** unattended, producing the win-back's first ever revenue at **~2.7x the reach** once I proved its eligibility logic had excluded most of its audience
- **Made a children's app safe from notification injection** — anyone with a user ID could push arbitrary content until I bearer-authed four public endpoints, **zero caller changes**

Software Engineer Intern, BitMango (Puzzle1 Studio)

June 2025 – Aug 2025 · Pangyo, South Korea

- **Kept 50+ live Unity mobile titles shipping** across a summer of releases, holding UI/UX and gameplay stable
- **Cleared 100+ QA-reported bugs into production**, in Unity prefabs and C# across a portfolio no single engineer owned
- **Kept the portfolio compliant and in the stores** through a platform requirement change, upgrading SDKs fleet-wide

Project Lead & Product Designer, Penn Spark

Jan 2025 – Present · Philadelphia, PA

- **Shipped client-facing 0-to-1 products** leading a cross-functional team of **15+** designers and developers

Database Engineer Intern, IT-Farm

June 2022 – Aug 2022 · Seongnam, South Korea

- **Made semiconductor data from 20+ client companies queryable** — including Samsung, SK and LG — by designing a SQL database prototype managing **1M+ entities** in Oracle and SQLite
- **Cut retrieval time across large-scale datasets** by restructuring and optimizing the queries the team relied on

PROJECTS

Capsule | React, Three.js (R3F), Blender

Jan 2025 – Apr 2025

- **Made memories something you walk through rather than scroll** — the **3D gallery** in React Three Fiber for a collaborative time-capsule app (team of 4), with direct-manipulation placement
- **Owned how the product felt to use** across the create → contribute → unlock flow

Mini Maya | C++, OpenGL, Qt

Feb 2026 – Mar 2026

- **Made arbitrary meshes editable in place** — a **half-edge structure** giving topology-aware traversal, plus an OBJ parser accepting arbitrary n-gons, rendered via OpenGL VBOs
- **Turned coarse geometry into smooth surfaces** with **Catmull–Clark subdivision** and topology ops (split, extrusion, triangulation) in a Qt editor

Mini Minecraft | C++, OpenGL, GLSL

Apr 2026

- **Made a from-scratch voxel world read as varied rather than noisy** — **7-biome procedural generation** (Perlin/FBM/Voronoi) and **3D Perlin caves** in a C++/OpenGL engine (team of 3)
- **Made that world legible** — the GLSL pipeline I owned: PCF shadows, screen-space reflections, vertex AO, day-night cycle

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, C++, C#, SQL, HTML/CSS, OCaml

Frameworks & Libraries: React, Next.js, Node.js, Astro, Tailwind CSS, Three.js, React Three Fiber, JUnit

Developer Tools: Git, GitLab, AWS, MongoDB, SQLite, Oracle, Figma, VS Code, IntelliJ

Graphics & 3D: OpenGL, GLSL, Unreal Engine 5, Unity, Maya, Blender